The Galois-Closure Descent Family: the non-Galois CM shapes of $R2'$ (GST Hodge Campaign, Round 37)

GST campaign continuation session (operator order: the standing finish order, `research/gpt6-native-point-primitive-span-20261017`, commit after Round 36 $2c94b3c6$)

2026-11-05 (session clock)

Scope and the standing order

This round strikes the face (a') named by Round 36's Theorem S17(iii): the *non-Galois CM shapes* — every CM closure pair $(\widetilde{G}, H)$ beyond the abelian-cyclotomic worlds, at the committed census (degrees 4 and 6 complete, degree 8 the regular and automorphism-split order-16 families). The operator's standing order is the finish order against Question 2; per the closing discipline (§40) the Clay statement is **not** claimed here, and the polarity split stands. Everything below is decided by the certificate `galois_closure_descent_round37_checks.py` (1950 checks, 0 failures, exit 0, deterministic double-run byte-identical, md5 `50d0b69880eb96a2bff95afe7da92c52`, ≈134 s, exact arithmetic, zero floats; the post-repair state after AUDIT-37 — the pre-repair certificate stood at 1892 checks, md5 `e16164e5ae0632572db5f01c93025e29`). *Provenance*: the branch identifier and the Round-36 push hash in the byline are the receipts recorded in the campaign worklog (CX-16) and the Round-32 record; the round artifacts live under `hodge-campaign/round37/` on the research branch.

1 The closure letter model

1.1 Setup

Let K be a CM field of degree $2g$ with Galois closure $\widetilde{K}$, $G := \mathrm{Gal}(\widetilde{K}/\mathbb{Q})$ (order n), $H := \mathrm{Gal}(\widetilde{K}/K)$, and complex conjugation $\rho \in Z(G)$ acting freely on the letter set $X := G/H$ (the $2g$ embeddings $K \hookrightarrow \widetilde{K}$; freeness of ρ is the CM condition $\rho \notin \bigcup_g gHg^{-1}$). A *CM type* $\Phi \subseteq X$ picks one letter from each ρ-pair. Write $A(K,\Phi)$ for the abelian variety of dimension g with complex multiplication by $\mathcal{O}_K$ and type Φ, and $\mathcal{H}^{p,p}_\Phi$ for the span of the letter words $e_U = e_{u_1} \wedge \cdots \wedge e_{u_{2p}}$ ($U \subseteq X$, $|U| = 2p$, $|U \cap \Phi| = p$) in $\bigwedge^{2p}(\widetilde{K} \otimes H^1)$.

Lemma 1.1 (Identification)**.** *$H^{2p}(A(K,\Phi),\mathbb{Q}) \otimes_{\mathbb{Q}} \widetilde{K} \cong \bigoplus_U \widetilde{K}\, e_U$ compatibly with the G-action $g \cdot e_\sigma = e_{g\circ\sigma}$ on the letters and the Galois action on coefficients; consequently*

$$\mathrm{Hdg}^{2p}\big(A(K,\Phi)\big) \;=\; \Big\{\, c = (c_U) \in \widetilde{K}^{\,\mathcal{B}_\Phi} \;:\; c_{gU} = g(c_U)\, s(g,U) \;\;\forall g \in G \,\Big\},$$

where $\mathcal{B}_\Phi$ is the set of Φ-balanced *words ($|U \cap \Phi| = p$) and $s(g,U) \in \{\pm 1\}$ is the exterior sign of the permutation sorting $g(U)$.*

Proof. $H^1(A) \otimes \widetilde{K}$ is the direct sum of the K-eigenlines $\widetilde{K}\, e_\sigma$, $\sigma \in X$; the eigenline for σ is mapped by $1 \otimes g$ into the eigenline for $g \circ \sigma$. Choosing eigenvectors compatibly on each G-orbit of letters fixes the identification up to a scalar cocycle on the stabilizers; the cocycle multiplies the twisted action below by a unit but never enters a trace (Lemma 1.3), so the dimensions and the product ranks computed from any compatible choice agree. A rational class ζ has $\zeta \otimes 1 = \sum_U c_U e_U$; rationality is $(1 \otimes g)(\zeta \otimes 1) = \zeta \otimes 1$, i.e. the displayed equivariance; the Hodge condition (p, p) is support on $\mathcal{B}_\Phi$. □

Lemma 1.2 (Mixed-orbit kill)**.** *If U is Φ-balanced and gU is not, then every system of Lemma 1.1 has $c_U = 0$. Hence the support of any rational Hodge class lies in the set of* pure-balanced *words*

$$\mathcal{P}_\Phi(p) = \{U : |U| = 2p,\ |U \cap \Phi| = p,\ \textit{and}\ |gU \cap \Phi| = p\ \forall g \in G\}.$$

Proof. The equivariance at g reads $c_{gU} = g(c_U) s(g, U)$; the left side vanishes on components outside $\mathcal{B}_\Phi$ by the Hodge condition, so $g(c_U) = 0$ and $c_U = 0$. □

1.2 The full-descent dimension law

Lemma 1.3 (Trace law)**.** *For $h \in G$, $h \neq e$, the trace of h on $\widetilde{K}$ (as $\mathbb{Q}$-vector space) is 0; consequently for any subgroup $S \leq G$ and any character twist $\chi : S \to \{\pm 1\}$,*

$$\dim_{\mathbb{Q}} \{x \in \widetilde{K} : h(x) = \chi(h)\, x\ \forall h \in S\} = \frac{n}{|S|},$$

independently of χ.

Proof. By the normal basis theorem $\widetilde{K} \cong \mathbb{Q}[G]$ as a G-module, and left multiplication by $h \neq e$ is a fixed-point-free permutation of the basis. The fixed space of the twisted action is the χ-isotypic part of $\mathrm{Res}^G_S \mathbb{Q}[G]$, of dimension $\frac{1}{|S|} \sum_{h \in S} \chi(h) \operatorname{tr}(h) = \frac{n}{|S|}$ since $\operatorname{tr}(h) = 0$ for $h \neq e$ and $\chi(e) = 1$. □

Theorem 1.4 (S18, Galois-closure descent)**.** *Let O be a G-orbit of pure-balanced words with stabilizer $S = S(U)$. The systems of Lemma 1.1 supported on O form a $\mathbb{Q}$-space of dimension $|O|$; in particular*

$$\dim_{\mathbb{Q}} \mathrm{Hdg}^{2p}\big(A(K, \Phi)\big) \ = \ \#\mathcal{P}_\Phi(p).$$

Proof. A system on O is determined by $c_{U_0} \in \widetilde{K}$ through $c_{gU_0} = g(c_{U_0})\, s(g, U_0)$; well-definedness under $gU_0 = g'U_0$ is exactly the twisted-fixed condition for S on c_{U_0} with the character $s(\cdot, U_0)|_S$ (a homomorphism since S stabilizes U_0). Lemma 1.3 gives $\dim = n/|S| = |O|$. Summing over orbits gives the dimension law. The certificate verifies this closed form against an exact blockwise RREF solution of the equivariance system at every committed world, type orbit, and degree (bounded $\#\mathcal{P} \cdot n \leq 2600$): the two algorithms agree everywhere. □

1.3 The census

Degree 4 (complete). The 2-generation sweep of S_4 (all cyclic representatives paired with all elements, conjugacy-canonicalized) finds exactly 5 transitive classes, matching the classical count. The central fixed-point-free involution filter selects $\{V_4, C_4, D_4\}$: the abelian worlds (Round

36's $m = 8, 12$ and $m = 5$ shapes, now re-derived through the new engine) and the *non-Galois quartic* world D_4 ($|H| = 2$). A_4 and S_4 have trivial centers and are excluded.

Degree 6 (complete). The same sweep of S_6 finds exactly 16 transitive classes, matching the classical count: since every class found is 2-generated, a hypothetical 17th transitive subgroup would contradict the classical census, so the sweep is complete. The filter selects four ρ-worlds: the abelian C_6 (Round 36's $m = 7, 9$ shapes) and the non-abelian classes of orders 12 (D_6, the generic sextic CM closure), 24, and 48.

Degree 8 (committed family). The five regular worlds (all abstract groups of order 8: C_8, $C_4 \times C_2$, C_2^3, D_4, Q_8, pairwise non-isomorphic by involution counts 1/3/7/5/1 with C_8 vs Q_8 split by exponent) and the order-16 worlds $\langle N_{\mathrm{reg}}, \tilde{\alpha}\rangle$ for automorphisms α of each N (complete in the class of transitive order-16 subgroups *containing* a regular subgroup: any such G has $G \cap N_{\mathrm{reg}} = N$ and $G/N \cong C_2$, so $G = \langle N, \tilde{\alpha}\rangle$ inside the holomorph). Worlds with multiple central fixed-point-free involutions contribute one world per ρ. Total: 110 (G, ρ)-worlds.

1.4 The dimension laws at the census

Proposition 1.5 (Type classification and the classical laws)**.** *1. Φ is induced from the CM subfield $k = \widetilde{K}^{H'}$ ($H < H' < G$, $\rho \notin H'$) iff Φ is a union of right H'-orbits of letters (the fibers of $G/H \to G/H'$).*

2. At every primitive *type of every committed world (116 type orbits across the census),* $\dim \mathrm{Hdg}^{2p} = \binom{g}{p}$ *for all p; in particular* $\dim \mathrm{NS} \otimes \mathbb{Q} = g$ *(Rosati/Weil).*

3. At every type induced from a quadratic *CM subfield,* $\dim \mathrm{Hdg}^{2p} = \binom{g}{p}^2$ *(the split law; the E^g-worlds). The certificate checks this at every* quadratic-inducible *orbit — any index-2 stabilizing H', not only the maximal-subfield-2 ones: 128 orbits, including the order-refined octic split types whose maximal inducing subfield is quartic.*

4. The palindrome law $\dim(2p) = \dim(2g - 2p)$ *for* $1 \le p \le g - 1$ *and the top degree* $\dim(2g) = 1$ *hold at every type.*

*5. Every type on a V_4-Galois quartic is induced (for $\Phi = \{x, y\}$, the subgroup $\langle x^{-1}y\rangle$ stabilizes Φ since $(x^{-1}y)^2 = e$ in V_4 and $y \neq \rho x$); likewise every type on a D_4-*regular *octic is induced from one of the four quartic CM subfields. Hence V_4-Galois quartics and D_4-Galois octics admit no simple CM abelian varieties; the computed values* $\dim \mathrm{NS} \in \{4\}$ *(quartic) and* $\{8\}$ *(octic) are the product-world values $B \times B'$.*

6. The Q_8-regular octic has only primitive types (its order-4 subgroups all contain ρ, so it has no proper CM subfield), with the binomial row $[4, 6, 4, 1]$.

Round 36's receipts are reproduced through the new engine at the abelian worlds: the quartic span 1, the sextic 3/9, and the octic 6/18/36 rows (primitive/intermediate/split).

2 The pairing-generation theorem

Theorem 2.1 (S19, non-Galois pairing generation)**.** *At every committed world (G, ρ) of the census (§1.3), every CM type orbit, and every degree $2p \le 2g$: every rational Hodge class on $A(K, \Phi)$ is a $\mathbb{Q}$-combination of* products of divisor classes*. Equivalently*

$$\operatorname{rank} \operatorname{span}\{d_{i_1} \wedge \cdots \wedge d_{i_p}\} = \dim_{\mathbb{Q}} \mathrm{Hdg}^{2p}\big(A(K, \Phi)\big), \qquad \textit{SURPLUS} = 0,$$

*the products taken with repetition (the diagonal $d \wedge d \neq 0$ for non-decomposable d; the iterate-basis sampler $S_p = \mathrm{span}(S_{p-1} \wedge \mathrm{NS})$ includes them by construction). The certificate computes **1386** (world, type, degree) surplus entries: **0 are nonzero**.*

Proof. The cup-product algebra of the complex torus $A(\mathbb{C})$ is the exterior algebra of H^1, so products of degree-2 classes are exterior products of the corresponding systems; products of equivariant systems are equivariant (cocycle multiplicativity), and products of balanced words are balanced, so the products land in the Hodge space. The span rank is computed exactly by integer rank maintenance over the pure-word coordinates; the certificate additionally verifies (i) the product-equivariance relation on explicit products, (ii) the random-class nondegeneracy $v^{\wedge p} \neq 0$ at every world (deterministic xorshift64* draws, with bounded retries — a single draw can be degenerate), and (iii) the calibration battery of Proposition 1.5. □

Corollary 2.2 (S19, rational Hodge conjecture at the census)**.** *The rational Hodge conjecture holds, at every degree, on every CM abelian variety whose closure pair $(\widetilde{G}, H)$ lies in the committed census — simple (the primitive type orbits, including all non-Galois primitive shapes at degrees 4 and 6 and the committed degree-8 families), imprimitive (the split and intermediate product worlds), non-abelian-Galois (the D_4-, Q_8-regular and automorphism-split order-16 octics), and on $E \times A$ over them (63 joint frames over 17 base worlds: the complete quartic and sextic census worlds, the five regular octic worlds, and three distinct automorphism-split order-16 bases C_8, $C_4 \times C_2$, C_2^3; per frame the first two type-orbit representatives — the disjoint frames calibrated by the tensor law* $\dim = 1 + \mathrm{NS}(A)$*, the coupled frames by the fiber rule — the full-fiber types carry* $\dim \mathrm{Hom}(E, A) = 2|\Phi_A|$ *and the E^{g+1}-law, the straddling types carry* $\dim \mathrm{Hom} = 0$*: no elliptic subvarieties in the simple worlds, distinct CM fields in the split ones).*

3 The re-drawn boundary (S20)

Theorem 3.1 (S20)**.** *(i) The elliptic face of R2′ is closed (Round 35); the abelian non-elliptic face at the abelian-cyclotomic shapes is closed (Round 36);* ***the non-Galois / non-abelian-Galois face is now closed at the committed census*** *— the merged-sector surplus at these shapes is* zero *at every degree, type, and arity covered.*

(ii) The residual of (a′) beyond the census: degree-8 closure groups of order ≥ 24 or without regular subgroups, and degrees ≥ 10 — the family-transport law (§45) carries the statement shape; no claim.

(iii) The remaining face of R2′: (b) the non-split Weil fourfold world ($\mathrm{NS}(A)_{\mathbb{Q}} = \mathbb{Q}\theta$, the 2-dimensional Weil plane, the committed obstruction receipts of Round 34).

(iv) R3 (the K-condition at general input beyond the channels) and R4 (the transcendental fold-birth with the (p, p)-sensitive gate) stand untouched.

The certificate

`galois_closure_descent_round37_checks.py`: V1 the census (completeness receipts 5/5 and 16/16 against the classical counts; the degree-8 family as in §1.3); V2 the model at every world (exterior cocycle law verified on all (g, h, U) at degrees 2 and 4; the mixed-orbit kill live at 224

type-orbits; the two-algorithm dimension agreement at all 110 worlds); V3 the crux (the classifier, the calibration battery of Proposition 1.5, SURPLUS= 0 at 1386 entries, the random-class and product-equivariance receipts); V4 the $E \times A$ frames; V5 the operator's calibration anchor (arity totals 4/8/24/64/208 by two algorithms at every $m \leq 5$, the $m = 6$ control 688 by both, $\det(I + C_L) = 1 - (-1)^L$ by exact Bareiss for $L = 2..12$); V6 the hostile battery (basis-deletion, wrong-sign cocycle flip, the type-validity battery — double-letter, missing-pair, duplicate-letter, empty, at a canary world and adversarially at every census world — the unbalanced-word injection into the filter's input list, garbage liveness, the census pair-generation and generator-deletion — every control fires with the predicted failure) plus the mechanical vacuity and self-comparison scans. Two defects were caught and repaired during authoring by the certificate's own calibrations (the coupled-frame closure late-binding and the single-draw degeneracy); both are documented in the file. AUDIT-37 then found and repaired thirteen further findings in one batched round (the S3b one-directional-canary gap closed by the adversarial type battery; the split-law domain widened to all 128 quadratic-inducible orbits; the E×A frame census extended to 17 base worlds; the degree-2 spans made genuine; the H4/H6 controls made load-bearing; the Lean PART-A label corrected); the closing re-check stands at 1950/0/exit 0, deterministic.

Boundary on every artifact

Q1 (the framework conjecture on the committed world model) remains proven and certified, unchanged. Q2 (the Clay statement) is **not claimed**; the closing string is **not emitted**; the polarity split stands. The falsifiable line after this round: (b) the non-split Weil fourfold; $R3$; $R4$; the transport family of (a') beyond the census.